The ∞-Categorical Yoneda Lemma

Note to the Reader

The result proved below is a formal consequence of the straightening– unstraightening equivalence, the universal property of sections of a left fibration, and the finality of the terminal object of a slice ∞-category. We work throughout in the quasicategorical model and assume familiarity with the straightening formalism of *Higher Topos Theory*, §2.2.1, and the finality criterion of *Higher Topos Theory*, §4.1.3.1.

1 Conventions

We work in the quasicategorical model of ∞-categories. Thus an ∞-category is a simplicial set satisfying the inner horn-filling condition. Equivalences are equivalences in the sense of ∞-categories; limits are limits in the ambient ∞-categorical sense, hence homotopy limits; mapping spaces are the canonical Kan complexes attached to an ∞-category. Fix Grothendieck universes $\mathcal{U} \in \mathcal{V}$. The adjective "small" means "$\mathcal{U}$-small." The ∞-category $\mathcal{S}$ of spaces is taken in $\mathcal{V}$, with the full subcategory of $\mathcal{U}$-small spaces understood implicitly. Consequently, if $\mathcal{C}$ is a small ∞-category, then

$$\mathcal{P}(\mathcal{C}) := \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})$$

is a $\mathcal{V}$-small ∞-category.

For an ∞-category K, we write $\mathrm{LFib}(K)$ for the full subcategory of $\mathrm{Cat}_{\infty,/K}$ spanned by left fibrations over K. If $p\colon X \to K$ is a left fibration and $k \in K$ is an object, its fiber at k means the pullback

$$X_k := X \times_K \{k\},$$

regarded as a space. If $r\colon X \to K$ is any morphism in Cat_∞, we write

$$\Gamma(r) := \mathrm{Map}_{\mathrm{Cat}_{\infty,/K}}(\mathrm{id}_K, r)$$

for the space of sections of r.

We fix once and for all the straightening–unstraightening equivalence in the form

$$\mathrm{St}_K\colon \mathrm{LFib}(K) \xrightarrow{\simeq} \mathrm{Fun}(K, \mathcal{S}), \qquad \mathrm{Un}_K\colon \mathrm{Fun}(K, \mathcal{S}) \xrightarrow{\simeq} \mathrm{LFib}(K),$$

functorial in K; cf. HTT, §2.2.1. All identifications below are taken with respect to this equivalence and the universal properties it exhibits.

2 Representable left fibrations

Let $\mathcal{C}$ be a small ∞-category. For $C \in \mathcal{C}$, define the representable presheaf

$$j(C) := \mathrm{Map}_{\mathcal{C}}(-, C) \in \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}).$$

This construction is functorial in C and determines the Yoneda embedding

$$j\colon \mathcal{C} \to \mathcal{P}(\mathcal{C}).$$

The geometric content of representability is the following standard fact.

Proposition 2.1. *For each object $C \in \mathcal{C}$, the projection*

$$q_C\colon \mathcal{C}^{\mathrm{op}}_{/C} \to \mathcal{C}^{\mathrm{op}}$$

is a left fibration, and there is a canonical equivalence

$$\mathrm{St}_{\mathcal{C}^{\mathrm{op}}}(q_C) \simeq j(C).$$

Moreover, the object $\mathrm{id}_C \in \mathcal{C}^{\mathrm{op}}_{/C}$ is terminal: for every object $x \in \mathcal{C}^{\mathrm{op}}_{/C}$, the mapping space

$$\mathrm{Map}_{\mathcal{C}^{\mathrm{op}}_{/C}}(x, \mathrm{id}_C)$$

is contractible.

Proof. The first assertion is the representability statement attached to straightening–unstraightening for slice left fibrations; equivalently, for each object $D \in \mathcal{C}$, the fiber of q_C at the corresponding object of $\mathcal{C}^{\mathrm{op}}$ is canonically equivalent to $\mathrm{Map}_{\mathcal{C}}(D, C)$, compatibly with transport along edges of $\mathcal{C}^{\mathrm{op}}$. This identifies $\mathrm{St}_{\mathcal{C}^{\mathrm{op}}}(q_C)$ with $j(C)$.

For terminality, let $x \in \mathcal{C}^{\mathrm{op}}_{/C}$. By the definition of the slice ∞-category, the mapping space from x to id_C is the fiber of the map from the space of 2-simplices of $\mathcal{C}$ with terminal vertex C to the space of pairs of composable edges having composite the edge corresponding to x, taken at the degenerate factorization of that edge through id_C. This fiber is contractible. Hence $\mathrm{Map}_{\mathcal{C}^{\mathrm{op}}_{/C}}(x, \mathrm{id}_C)$ is contractible. □

3 Sections and limits

The Yoneda lemma will be obtained from two formal identifications.

Proposition 3.1. *Let $q\colon X \to K$ and $p\colon Y \to K$ be morphisms in Cat_∞, and let*

$$\begin{tikzcd} Y \times_K X \arrow[r] \arrow[d, "p_X"'] & Y \arrow[d, "p"] \\ X \arrow[r, "q"'] & K \end{tikzcd}$$

be a pullback square. Then the universal property of the pullback induces a canonical equivalence of spaces

$$\mathrm{Map}_{\mathrm{Cat}_{\infty,/K}}(q, p) \xrightarrow{\simeq} \Gamma(p_X).$$

If p is a left fibration, then so is p_X.

Proof. A point of $\mathrm{Map}_{\mathrm{Cat}_{\infty,/K}}(q,p)$ is, by definition, a morphism $\alpha\colon X \to Y$ in Cat_∞ together with an equivalence $p \circ \alpha \simeq q$ in $\mathrm{Fun}(X,K)$. By the universal property of the pullback, such data are equivalent to a morphism $\widetilde{\alpha}\colon X \to Y \times_K X$ over X, that is, to a section of p_X. This yields the displayed equivalence. Stability of left fibrations under pullback is standard. □

Proposition 3.2. *Let K be an ∞-category, and let $r\colon X \to K$ be a left fibration classified by a functor $G\colon K \to \mathcal{S}$. Then there is a canonical equivalence*

$$\Gamma(r) \xrightarrow{\simeq} \lim_K G,$$

functorial in G.

Proof. Under straightening, the left fibration r corresponds to G. Since straightening is an equivalence of ∞-categories, it identifies the mapping space

$$\Gamma(r) = \mathrm{Map}_{\mathrm{Cat}_{\infty,/K}}(\mathrm{id}_K, r)$$

with the mapping space in $\mathrm{Fun}(K,\mathcal{S})$ from the terminal object to G. By the universal property of the limit, the latter mapping space is canonically identified with $\lim_K G$. Functoriality is inherited from functoriality of straightening and the universal property of the limit functor. □

4 Finality in the slice

Fix an object $C \in \mathcal{C}$. Let

$$i_C\colon \{\mathrm{id}_C\} \hookrightarrow \mathcal{C}^{\mathrm{op}}_{/C}$$

denote the inclusion of the full simplicial subset spanned by the terminal object.

Proposition 4.1. *The functor i_C is final.*

Proof. By HTT, §4.1.3.1, it suffices to show that for each object $x \in \mathcal{C}^{\mathrm{op}}_{/C}$, the simplicial set

$$\{\mathrm{id}_C\} \times_{\mathcal{C}^{\mathrm{op}}_{/C}} (\mathcal{C}^{\mathrm{op}}_{/C})_{x/}$$

is weakly contractible. By the defining universal property of the overcategory, this pullback is canonically equivalent to the mapping space

$$\mathrm{Map}_{\mathcal{C}^{\mathrm{op}}_{/C}}(x, \mathrm{id}_C).$$

The latter is contractible by Proposition 2.1. Hence i_C is final. □

Corollary 4.2. *For every functor $G\colon \mathcal{C}^{\mathrm{op}}_{/C} \to \mathcal{S}$, restriction along i_C induces a canonical equivalence*

$$\lim_{\mathcal{C}^{\mathrm{op}}_{/C}} G \xrightarrow{\simeq} G(\mathrm{id}_C).$$

Proof. Since i_C is final, restriction along i_C induces an equivalence

$$\lim_{\mathcal{C}^{\mathrm{op}}_{/C}} G \xrightarrow{\simeq} \lim_{\{\mathrm{id}_C\}} G|_{\{\mathrm{id}_C\}}.$$

The indexing ∞-category on the right is contractible with one object, so its limit is canonically equivalent to the value at that object. □

5 Yoneda

The Yoneda lemma is now immediate from the preceding infrastructure.

Theorem 5.1 (Yoneda lemma)**.** *The bifunctor*

$$\mathcal{C}^{\mathrm{op}} \times \mathcal{P}(\mathcal{C}) \to \mathcal{S}, \qquad (C, F) \longmapsto \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(j(C), F),$$

is canonically equivalent to the evaluation bifunctor

$$\mathrm{ev}\colon \mathcal{C}^{\mathrm{op}} \times \mathcal{P}(\mathcal{C}) \to \mathcal{S}, \qquad (C, F) \longmapsto F(C).$$

Equivalently, for each $C \in \mathcal{C}$ and each $F \in \mathcal{P}(\mathcal{C})$ there is a canonical natural equivalence

$$\Phi_{C,F}\colon \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(j(C), F) \xrightarrow{\simeq} F(C).$$

Proof. Fix $(C, F) \in \mathcal{C}^{\mathrm{op}} \times \mathcal{P}(\mathcal{C})$. Let

$$p\colon \mathcal{E} \to \mathcal{C}^{\mathrm{op}}$$

be a left fibration classified by F, and let

$$q_C\colon \mathcal{C}^{\mathrm{op}}_{/C} \to \mathcal{C}^{\mathrm{op}}$$

be the representable left fibration classified by $j(C)$. Since $\mathrm{St}_{\mathcal{C}^{\mathrm{op}}}$ is an equivalence, it induces an equivalence on mapping spaces:

$$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(j(C), F) \simeq \mathrm{Map}_{\mathrm{LFib}(\mathcal{C}^{\mathrm{op}})}(q_C, p).$$

Because $\mathrm{LFib}(\mathcal{C}^{\mathrm{op}})$ is a full subcategory of $\mathrm{Cat}_{\infty,/\mathcal{C}^{\mathrm{op}}}$, the right-hand side may be computed in the ambient slice ∞-category.

Form the pullback square

$$\begin{tikzcd} \mathcal{E}' \arrow[r] \arrow[d, "p'"'] & \mathcal{E} \arrow[d, "p"] \\ \mathcal{C}^{\mathrm{op}}_{/C} \arrow[r, "q_C"'] & \mathcal{C}^{\mathrm{op}}. \end{tikzcd}$$

By Proposition 3.1, there is a canonical equivalence

$$\mathrm{Map}_{\mathrm{LFib}(\mathcal{C}^{\mathrm{op}})}(q_C, p) \simeq \Gamma(p').$$

Let

$$F'\colon \mathcal{C}^{\mathrm{op}}_{/C} \to \mathcal{S}$$

be the functor classified by p'. By Proposition 3.2,

$$\Gamma(p') \simeq \lim_{\mathcal{C}^{\mathrm{op}}_{/C}} F'.$$

By Corollary 4.2, restriction along i_C yields a canonical equivalence

$$\lim_{\mathcal{C}^{\mathrm{op}}_{/C}} F' \simeq F'(\mathrm{id}_C).$$

Therefore

$$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(j(C), F) \simeq F'(\mathrm{id}_C).$$

It remains to identify $F'(\mathrm{id}_C)$ with $F(C)$. Since F' is the straightening of p', the universal property of straightening identifies $F'(\mathrm{id}_C)$ with the fiber $\mathcal{E}'_{\mathrm{id}_C}$. On the other hand, by the universal property of the pullback square defining $\mathcal{E}'$, the fiber of p' at id_C is canonically equivalent to the fiber of p at the image of id_C under q_C, namely the object C of $\mathcal{C}^{\mathrm{op}}$. Thus there are canonical equivalences

$$F'(\mathrm{id}_C) \simeq \mathcal{E}'_{\mathrm{id}_C} \simeq \mathcal{E}_C.$$

Since p is classified by F, straightening identifies $\mathcal{E}_C$ with $F(C)$. Combining the preceding equivalences gives

$$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(j(C), F) \simeq F(C).$$

Each step is functorial in both variables: $C \mapsto q_C$ is functorial by the functoriality of the representable left fibration, $F \mapsto p$ is functorial by unstraightening, pullback is functorial in the diagram, the equivalence between sections and limits is functorial in the classified diagram, and restriction along the final functor i_C is functorial in the diagram. Hence the composite construction refines to an equivalence of bifunctors. ◻

Corollary 5.2. *The Yoneda embedding*

$$j \colon \mathcal{C} \to \mathcal{P}(\mathcal{C})$$

is fully faithful. Equivalently, for all objects $C, D \in \mathcal{C}$, the canonical map

$$\mathrm{Map}_{\mathcal{C}}(C, D) \to \mathrm{Map}_{\mathcal{P}(\mathcal{C})}(j(C), j(D))$$

is an equivalence of spaces.

Proof. Apply Theorem 5.1 with $F = j(D)$. One obtains

$$\mathrm{Map}_{\mathcal{P}(\mathcal{C})}(j(C), j(D)) \simeq j(D)(C) \simeq \mathrm{Map}_{\mathcal{C}}(C, D).$$

This equivalence is natural in (C, D), hence identifies the mapping-space bifunctor of $\mathcal{C}$ with that of the essential image of j. ◻

Remark 5.3. The theorem is not a calculation internal to the presheaf ∞-category. It is the assertion that the representable object $j(C)$ is geometrically realized by the slice left fibration $\mathcal{C}^{\mathrm{op}}_{/C} \to \mathcal{C}^{\mathrm{op}}$, and that the universal property of the terminal object id_C computes sections of its pullback against an arbitrary left fibration. The Yoneda equivalence is therefore the composite of straightening, pullback stability, finality, and the universal property of the limit.